

# UNIFORM SHEAFY TATE DOMAINS THAT ARE NOT STABLY UNIFORM

ANONYMOUS

**ABSTRACT.** Buzzard–Verberkmoes and Mihara show that a stably uniform Tate Huber ring is sheafy. We construct two uniform sheafy Tate domains which are not stably uniform, answering Question 7 of Kedlaya’s *Nonarchimedean Scottish Book*. In the first example a rational localisation is nonreduced, whilst in the second the nonuniform rational localisation is an integral domain. All the results in this paper are due to ChatGPT 5.6 Sol. Both constructions, including their sheafiness and the rational localisations witnessing failure of stable uniformity, have been formalised in Lean 4.

## 1. INTRODUCTION

The point at issue is the behaviour of rational localisations. The examples of Buzzard–Verberkmoes and Mihara show that a uniform Tate Huber ring can lose uniformity on a rational chart and, separately, that uniformity alone does not imply sheafiness [4, 13]. They leave open whether sheafiness itself prevents this loss of uniformity. Hansen and Kedlaya ask this explicitly [6, Remark 3.16]; it is also Question 7 of Kedlaya’s *Nonarchimedean Scottish Book* [14]. The answer is no.

All the results in this paper are due to ChatGPT 5.6 Sol. We have aimed to include references to related work that may have been used by ChatGPT in finding the examples; see remark 1.3.

We begin with what ChatGPT called the “finite-jet pullback”. This is a nonnoetherian uniform integral domain for which a distinguished rational localisation is nonreduced, and hence is no longer uniform. Let  $k$  be a complete discretely valued nonarchimedean field, with valuation ring  $k^\circ$ , uniformizer  $\varpi$ , and residue field  $\tilde{k}$ . Put

$$\begin{aligned} L_0 &= k^\circ \langle W, W^{-1} \rangle, & B_0 &= k^\circ \langle W, Q \rangle / (Q^2), \\ C_0 &= L_0 \langle Q \rangle, & D_0 &= L_0 \langle Q \rangle / (Q^2), \end{aligned}$$

where  $k^\circ \langle W, W^{-1} \rangle$  means  $k^\circ \langle W, V \rangle / (WV - 1)$ . There are natural maps  $B_0 \rightarrow D_0$  and  $C_0 \twoheadrightarrow D_0$ . Define

$$A_0 = B_0 \times_{D_0} C_0, \quad A = A_0[1/\varpi],$$

and endow  $A$  with the topology defined by the ring of definition  $A_0$ .

The main result is the following.

**Theorem 1.1.** *The ring  $A$  is a complete uniform nonnoetherian Tate  $k$ -algebra and an integral domain, with  $A^\circ = A_0$ . The Huber pair  $(A, A^\circ)$  is sheafy. However,*

$$A \left\langle \frac{W}{\varpi} \right\rangle \cong k \langle X, Q \rangle / (Q^2), \quad X = \frac{W}{\varpi},$$

so  $A$  is not stably uniform.

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2020 *Mathematics Subject Classification.* Primary 14G22; Secondary 13J10.

*Key words and phrases.* Huber ring, adic spectrum, sheafiness, stable uniformity, rational localisation, Milnor square.

The proof uses Milnor squares of complete Tate rings. Recall that a cartesian square of rings with a surjective leg is called a Milnor square [12, Section 2]; the complete Tate setting is used by Kerz–Saito–Tamme in their work on analytic  $K$ -theory [9, Section 3].

**Definition 1.2.** A commutative square of complete Tate  $k$ -algebras

$$\begin{array}{ccc} R & \longrightarrow & C \\ \downarrow & & \downarrow \\ B & \longrightarrow & D \end{array}$$

is a *strict Milnor square* if it is cartesian, the sequence of underlying topological  $k$ -vector spaces

$$0 \longrightarrow R \longrightarrow B \oplus C \xrightarrow{(b,c) \mapsto \bar{b} - \bar{c}} D \longrightarrow 0$$

is strict exact, and  $C \rightarrow D$  is a strict surjection.

After rotating the square if necessary, the initial Milnor square is strict by the nonarchimedean open mapping theorem [9, Lemma 3.1]. Our main input is that, for this finite-jet square, the strict exact row persists after completed rational localisation and is compatible with refinement. This transfers sheafiness from  $B, C, D$  to  $A$ . The second example shows that the failure of stable uniformity need not come from a nonreduced localisation.

The construction, uniformity, sheafiness, and failure of stable uniformity for both examples have been formalised in Lean 4. The formalisation also includes the rational-chart calculations used above. Appendix A records the relevant definitions and theorem statements.

**Remark 1.3** (Relation with earlier work). Buzzard–Verberkmoes construct a uniform affinoid ring with a nonuniform rational localisation and, separately, a uniform affinoid ring whose structure presheaf is not a sheaf [4, Propositions 17 and 18]. Mihara gives further examples of both phenomena [13]. These examples do not give a uniform sheafy ring which is not stably uniform.

Ben-Bassat–Kreminizer prove the corresponding one-variable strict resolutions for Weierstrass and Laurent localisations, and reduce a general rational localisation to these cases [2, Lemmas 5.13–5.14]. They do not formulate a multivariable Koszul theorem. That formulation, and the strict Koszul regularity of rational localisations, is given by Bambozzi–Kreminizer [3, Notation 4.3, Definitions 4.5–4.6, Proposition 4.9, Lemma 4.11, and Corollary 4.13]. The lattice estimates in lemma 5.1 are the form needed to show that rational localisation preserves the finite-jet Milnor square. The resulting uniform sheafy domain which is not stably uniform appears to be new.

The weighted-parity construction of section 7 is closer to the infinite monomial-support examples of Buzzard–Verberkmoes and Mihara. Its additional feature is that rational localisation can be calculated in a noetherian finite head.

## 2. CONVENTIONS AND NOTATION

We follow Huber for Huber pairs, adic spectra, rational subsets, and rational localisation [7], and use Wedhorn’s account for strong noetherianity and sheafiness [16, Proposition and Definition 6.36, Section 8.1]; the bounded-denominator language for strictness follows Buzzard–Verberkmoes [4], and the quotient presentation of a rational localisation is also discussed by Hansen–Kedlaya [6].

- *Tate topology.* Throughout,  $E$  denotes a complete Tate  $k$ -algebra. In the discretely valued setting of this paper, this means that  $E$  contains a topologically nilpotent unit and admits an open bounded  $k^\circ$ -subalgebra  $E_0$  such that

$$E = E_0[1/\varpi], \quad \{\varpi^n E_0\}_{n \geq 0}$$

is a basis of neighbourhoods of zero. Such an  $E_0$  is called a *ring of definition*. We always choose it closed, hence  $\varpi$ -adically complete. A subset of  $E$  is bounded if it is contained in  $\varpi^{-r}E_0$  for some  $r \geq 0$ . If  $M$  is a complete topological  $k$ -vector space, an open bounded  $k^\circ$ -submodule  $M_0 \subset M$  will be called a lattice.

- *Uniformity and noetherianity.* We write  $E^\circ$  for the ring of power-bounded elements. The ring  $E$  is *uniform* if  $E^\circ$  is bounded, and *strongly noetherian* if  $E\langle Z_1, \dots, Z_s \rangle$  is noetherian for every  $s \geq 0$ . A Tate ring is *stably uniform* if every rational localisation is uniform.
- *Huber pairs and sheafiness.* A *ring of integral elements* is an open subring  $E^+ \subseteq E^\circ$  which is integrally closed in  $E$ . We call the Huber pair  $(E, E^+)$  *sheafy* when its structure presheaf is a sheaf of complete topological rings; in particular, the map to the matching-family equaliser for a finite rational cover is a homeomorphism.
- *Strictness.* We use the following bounded-denominator form of strictness. Let  $u : M \rightarrow N$  be a continuous  $k$ -linear map and choose lattices  $M_0 \subset M$  and  $N_0 \subset N$ . Then  $u$  is strict if and only if there exists  $a \geq 0$  such that

$$(1) \quad \varpi^a(u(M) \cap N_0) \subset u(M_0).$$

For a surjection this becomes  $\varpi^a N_0 \subset u(M_0)$ . For an injection it says that the lattice induced from the target is bounded in the source. This is the criterion used by Buzzard–Verberkmoes [4, Lemma 2]. We use the term *strict Milnor square* in the sense of definition 1.2; the underlying algebraic terminology is standard [12, 9].

- *Rational localisation.* We take every rational datum to have at least one numerator; an empty list may be padded by  $f_1 = 0$ . Let  $\alpha = (f_1, \dots, f_m; g)$  be a rational datum in  $E$ , so the ideal generated by  $g, f_1, \dots, f_m$  is open. Since  $E$  is a Tate  $k$ -algebra, this ideal is the unit ideal. After multiplying the entire datum by a common power of  $\varpi$ , we may assume that all entries lie in  $E_0$ .

The rational localisation  $E_\alpha$  is the separated  $\varpi$ -adic completion of  $E[1/g]$  for the ring of definition generated by

$$E_0 \left[ \frac{f_1}{g}, \dots, \frac{f_m}{g} \right].$$

It is independent of the ring of definition and of the presentation of the rational domain, and it is transitive under rational refinement. Put

$$T_E = E\langle T_1, \dots, T_m \rangle = E_0\langle T_1, \dots, T_m \rangle[1/\varpi], \quad r_i = gT_i - f_i,$$

and

$$(2) \quad I_{E,\alpha} := (r_1, \dots, r_m) = \operatorname{im} \left( T_E^m \xrightarrow{d_{1,E}} T_E \right) \subset T_E, \quad d_{1,E}(u_1, \dots, u_m) = \sum_i u_i(gT_i - f_i),$$

Then

$$(3) \quad E_\alpha \cong T_E / \overline{I_{E,\alpha}}.$$

Here the bar is necessary until the ideal has been shown to be closed. This is the standard quotient presentation of rational localisation; see [7, (1.2), Proposition 1.3, Lemma 1.5] and [6, Definition 3.6].

## 3. THE GLOBAL RING

In this section we identify the finite-jet algebra with a concrete subring of  $C$  and prove that it is a complete uniform domain. Recall that

$$L_0 = k^\circ\langle W, W^{-1} \rangle, \quad B_0 = k^\circ\langle W, Q \rangle / (Q^2),$$

$$C_0 = L_0\langle Q \rangle, \quad D_0 = L_0\langle Q \rangle / (Q^2),$$

where  $k^\circ\langle W, W^{-1} \rangle$  denotes  $k^\circ\langle W, V \rangle / (WV - 1)$ . The natural maps  $B_0 \rightarrow D_0$  and  $C_0 \rightarrow D_0$  define

$$A_0 = B_0 \times_{D_0} C_0.$$

We write

$$A = A_0[1/\varpi], \quad L = L_0[1/\varpi], \quad B = B_0[1/\varpi], \quad C = C_0[1/\varpi], \quad D = D_0[1/\varpi].$$

Each of these Tate rings carries the topology defined by its subscript-zero ring.

The map  $C_0 \rightarrow D_0$  has a continuous  $k^\circ$ -linear section given by truncation in the  $Q$ -variable:

$$f_0 + Qf_1 \mapsto f_0 + Qf_1.$$

Consequently

$$(4) \quad 0 \longrightarrow A_0 \longrightarrow B_0 \oplus C_0 \longrightarrow D_0 \longrightarrow 0$$

is exact, and  $A_0$  is  $\varpi$ -adically complete. After inverting  $\varpi$  we obtain a strict Milnor square

$$\begin{array}{ccc} A & \longrightarrow & C \\ \downarrow & & \downarrow \\ B & \longrightarrow & D. \end{array}$$

In fact, the chosen rings of definition already make the integral row exact, so no powers of  $\varpi$  are lost in the defining square.

The map  $B_0 \rightarrow D_0$  is injective. Projection to  $C_0$  therefore identifies  $A_0$  with the subring

$$(5) \quad A_0 = \{f_0(W) + Qf_1(W) + Q^2h(W, W^{-1}, Q) : f_0, f_1 \in k^\circ\langle W \rangle, h \in C_0\}.$$

Equivalently,  $A_0$  consists of the restricted series

$$(6) \quad \sum_{(a,b) \in S} c_{a,b} W^a Q^b, \quad S = \{(a,b) \in \mathbf{Z} \times \mathbf{N} : b \leq 1 \Rightarrow a \geq 0\},$$

with  $c_{a,b} \in k^\circ$ . Here restricted means that, for every  $n$ , only finitely many coefficients are nonzero modulo  $\varpi^n$ . The set  $S$  is closed under addition, so this is a subring of  $C_0$ . The same description with coefficients in  $k$  gives  $A \subset C$ .

Equip  $A, B, C, D$  with the coefficientwise Gauss norms whose unit balls are the displayed rings of definition. Write

$$\iota_B : A \rightarrow B, \quad \iota_C : A \rightarrow C, \quad \rho_B : B \rightarrow D, \quad \rho_C : C \rightarrow D$$

for the structural maps, and give  $A$  the restricted Gauss norm from  $C$ . For  $a = f_0 + Qf_1 + Q^2h$ , the coefficient formulas give

$$(7) \quad \|\iota_C(a)\|_C = \|a\|_A, \quad \|\iota_B(a)\|_B \leq \|a\|_A, \quad \|\rho_B(b)\|_D \leq \|b\|_B, \quad \|\rho_C(c)\|_D \leq \|c\|_C.$$

Thus every structural map is norm-nonincreasing.

**Proposition 3.1.** *The ring  $A$  is a complete uniform nonnoetherian Tate  $k$ -algebra and an integral domain. Moreover,*

$$A^\circ = A_0.$$

*Proof.* The Gauss norm on  $L = k\langle W, W^{-1} \rangle$  is multiplicative, and the restricted Gauss norm on  $C = L\langle Q \rangle$  is therefore multiplicative as well. In particular,  $C$  is a domain, and hence so is its subring  $A$ . The conditions on the constant and linear  $Q$ -coefficients which define  $A$  are closed conditions in  $C$ , so  $A$  is complete. The image of  $\varpi$  in  $A$  is a topologically nilpotent unit, and therefore  $A$  is a Tate ring.

The coefficient description of  $A_0$  shows that it is exactly the unit ball for the restricted Gauss norm on  $A$ . Since this norm is multiplicative,

$$\|a^n\| = \|a\|^n.$$

It follows that  $a$  is power-bounded if and only if  $\|a\| \leq 1$ . Thus  $A^\circ = A_0$ , and the unit ball is bounded, so  $A$  is uniform.

For nonnoetherianity, let  $J = \ker(\iota_B) = Q^2C \subset A$ . If  $J$  were generated by  $a_1, \dots, a_r$ , then, for every  $\ell \in L$ , comparison of the  $Q^2$ -coefficient in an expression

$$Q^2\ell = \sum_i x_i a_i, \quad x_i \in A,$$

would express  $\ell$  as a  $k\langle W \rangle$ -linear combination of the  $Q^2$ -coefficients of the  $a_i$ . This would make  $L = k\langle W, W^{-1} \rangle$  a finite  $k\langle W \rangle$ -module, which is impossible: a finite algebra is integral, whereas a monic equation for  $W^{-1}$ , after multiplication by a power of  $W$ , would put 1 in the proper ideal  $(W) \subset k\langle W \rangle$ . Thus  $J$  is not finitely generated, and  $A$  is not noetherian.  $\square$

#### 4. A NONUNIFORM RATIONAL LOCALISATION

We next compute the rational chart which witnesses the failure of stable uniformity.

Consider the datum

$$\alpha_{\text{in}} = (W; \varpi).$$

The ideal  $(\varpi, W)$  is open, since  $\varpi$  is a unit in  $A$ . Let

$$A_{\text{in}} = A \left\langle \frac{W}{\varpi} \right\rangle$$

be the corresponding rational localisation on  $|W| \leq |\varpi|$ , and write  $X = W/\varpi$ . By the ring-of-definition construction,  $A_{\text{in}}$  is the separated completion of the subring generated by  $A_0$  and  $X$ .

**Proposition 4.1.** *There is a canonical isomorphism of complete topological rings*

$$A_{\text{in}} \cong k\langle X, Q \rangle / (Q^2), \quad W \mapsto \varpi X, \quad Q \mapsto Q.$$

*Proof.* Put

$$T = k\langle X, Q \rangle / (Q^2).$$

Take the two-jet map  $A \rightarrow B$  and then rescale its disc variable by  $W \mapsto \varpi X$ . This gives a continuous map

$$\theta : A \longrightarrow T, \quad f_0(W) + Qf_1(W) + Q^2h \longmapsto f_0(\varpi X) + Qf_1(\varpi X).$$

The element  $\theta(\varpi)$  is a unit and  $\theta(W)/\theta(\varpi) = X$  is power-bounded. The universal property first extends  $\theta$  to the algebraic rational localisation and then, by continuity, to a map

$$\Phi : A_{\text{in}} \longrightarrow T.$$

We next record why the two-jet map loses no information other than the  $Q^2$ -tail after completion. If the constant and linear  $Q$ -coefficients of  $y \in A$  vanish, then  $y_n = W^{-n}y$  still belongs to  $A$ , and the  $y_n$  have bounded norm. In  $A_{\text{in}}$  one has

$$\bar{y} = \varpi^n X^n \overline{y_n} \quad (n \geq 0).$$

The powers of  $X$  are bounded and  $\varpi^n \rightarrow 0$ , so  $\bar{y} = 0$ . In particular, the image  $\bar{Q}$  of  $Q$  satisfies  $\bar{Q}^2 = 0$ .

Let  $\bar{X} = \bar{W}/\varpi$ . Since  $\bar{X}$  is power-bounded, evaluation at  $\bar{X}$  gives a continuous map  $k\langle X \rangle \rightarrow A_{\text{in}}$ . We may therefore define

$$\Psi : T \longrightarrow A_{\text{in}}, \quad f + Qg \longmapsto f(\bar{X}) + g(\bar{X})\bar{Q}.$$

To check the first composite, decompose  $a \in A$  as  $a = f_0(W) + Qf_1(W) + y$ , where the first two coefficients lie in  $k\langle W \rangle$  and the first two  $Q$ -coefficients of  $y$  vanish. The preceding paragraph kills  $y$ , and hence  $\Psi(\theta(a)) = \bar{a}$ . This proves  $\Psi\Phi = 1$  on the dense algebraic rational localisation, and therefore on  $A_{\text{in}}$ . In the other direction,  $\Phi\Psi$  fixes constants,  $X$  and  $Q$ . It is consequently the identity first on polynomials, then on  $k\langle X \rangle$  by density, and finally on  $T$ . Thus  $\Phi$  and  $\Psi$  are inverse continuous homomorphisms.  $\square$

**Proposition 4.2.** *The ring  $A$  is not stably uniform.*

*Proof.* In  $T = k\langle X, Q \rangle / (Q^2)$ , every element of the line  $kQ$  is nilpotent and hence power-bounded. This line is not bounded with respect to the ring of definition  $T_0$ : for every  $r \geq 0$ ,

$$\varpi^r(\varpi^{-(r+1)}Q) = \varpi^{-1}Q \notin T_0.$$

Thus  $T$  is not uniform. By proposition 4.1, a rational localisation of the uniform ring  $A$  is nonuniform.  $\square$

## 5. LOCALISATION OF THE MILNOR SQUARE

In this section we prove that rational localisation preserves the defining pullback. We first record the strict Koszul estimate needed for the proof.

**5.1. Koszul complexes for rational localisations.** Let  $E$  be one of  $B, C, D$ , with its coefficientwise Gauss norm and the displayed unit ball  $E_0$ . Let  $f_1, \dots, f_m, g \in E_0$  generate the unit ideal in  $E$ . Put

$$T_E = E\langle T_1, \dots, T_m \rangle, \quad T_{E,0} = E_0\langle T_1, \dots, T_m \rangle,$$

and set  $r_i = gT_i - f_i$ .

We give  $T_E$  the topology for which  $\{\varpi^n T_{E,0}\}_{n \geq 0}$  is a basis of neighbourhoods of zero. The term  $\bigwedge^q T_E^m$  has the finite-module topology defined by the lattice  $\bigwedge^q T_{E,0}^m$ ; equivalently, in the standard exterior basis it has the finite product topology. Images have the subspace topology. Thus a differential is strict when it is open onto its image, or equivalently when its coimage is homeomorphic to its image.

Ben-Bassat–Kremnizer prove the one-variable strict two-term resolutions and then factor a general rational localisation into Laurent and Weierstrass localisations [2, Lemmas 5.13–5.14]; they do not state the multivariable Koszul assertion. Applied to  $E \in \{B, C, D\}$ , the strict Koszul regularity theorem of Bambozzi–Kremnizer [3, Notation 4.3, Definitions 4.5–4.6, Proposition 4.9, Lemma 4.11, and Corollary 4.13] gives the exactness, strictness, and closed-image assertions below. We include the direct proof followed in the formalisation because the two estimates relative to the displayed lattices are not stated there and are needed below.

**Lemma 5.1.** *The Koszul complex*

$$0 \longrightarrow \bigwedge^m T_E^m \longrightarrow \cdots \longrightarrow \bigwedge^2 T_E^m \xrightarrow{d_{2,E}} T_E^m \xrightarrow{d_{1,E}} T_E$$

is exact in positive degrees. Every differential is continuous and strict onto its image, and every image is closed. In particular, if

$$I_E = (r_1, \dots, r_m) = \text{im}(d_{1,E}) \subset T_E, \quad d_{1,E}(u_1, \dots, u_m) = \sum_i u_i r_i,$$

then  $I_E$  is closed. Moreover, there exists  $h_E \in \mathbf{Z}_{\geq 0}$  such that

$$(8) \quad \varpi^{h_E}(I_E \cap T_{E,0}) \subset d_{1,E}(T_{E,0}^m).$$

There also exists  $z_E \in \mathbf{Z}_{\geq 0}$  such that

$$(9) \quad \varpi^{z_E}(\ker(d_{1,E}) \cap T_{E,0}^m) \subset d_{2,E}(\bigwedge^2 T_{E,0}^m).$$

*Proof.* Over the polynomial ring  $E[T_1, \dots, T_m]$ , the sequence  $gT_i - f_i$  has Koszul complex exact in positive degrees. The map

$$E[T_1, \dots, T_m] \longrightarrow T_E$$

is flat: this is the usual identification of  $T_E$  with the  $\varpi$ -adic completion of  $E_0[T_1, \dots, T_m]$ , followed by inverting  $\varpi$ . Tensoring the polynomial Koszul complex with  $T_E$ , and using the standard base-change identification of its finite free terms, proves positive-degree exactness. Notice that this does not assert exactness in degree zero.

Each differential is continuous because its coordinates are finite sums of multiplication maps. The image of  $d_{1,E}$  is closed by the noetherian closed-ideal theorem. In higher degree, exactness identifies an image with the kernel of the next continuous differential, and hence that image is closed. The nonarchimedean open mapping theorem, applied to each differential with its closed image, now gives a constant  $C_q$  such that every element of the image has a preimage of norm at most  $C_q$  times its norm. This proves strictness.

For  $d_{1,E}$  choose  $h_E$  so that  $|\varpi|^{h_E} C \leq 1$ , where  $C$  is a controlled-lift constant for  $d_{1,E}$ . Scaling a controlled lift of an element of  $I_E \cap T_{E,0}$  gives (8). Applying the same argument to  $d_{2,E}$ , with its own constant and using  $\text{im}(d_{2,E}) = \ker(d_{1,E})$ , gives (9).  $\square$

**5.2. The defining ideals under pullback.** Fix a rational datum

$$\alpha = (f_1, \dots, f_m; g)$$

in  $A$ . Multiplying all entries by a common power of  $\varpi$ , we assume  $f_1, \dots, f_m, g \in A_0$ . For  $E \in \{A, B, C, D\}$  use the notation  $T_E$ ,  $T_{E,0}$ , and  $d_{1,E}$  introduced above, and write

$$I_E = (gT_1 - f_1, \dots, gT_m - f_m) \subset T_E.$$

The exact integral row (4) and the coefficientwise section of  $C_0 \rightarrow D_0$  give an exact sequence

$$(10) \quad 0 \longrightarrow T_{A,0} \longrightarrow T_{B,0} \oplus T_{C,0} \longrightarrow T_{D,0} \longrightarrow 0.$$

The map  $T_{C,0} \rightarrow T_{D,0}$  is surjective, as are the induced maps on finite free modules and their exterior powers. Exactness is coefficientwise: a compatible restricted coefficient pair lifts to  $A_0$ , and the lifted coefficients still tend to zero  $\varpi$ -adically.

**Lemma 5.2.** *The ideal  $I_A$  is closed in  $T_A$ , and the canonical map*

$$I_A \longrightarrow I_B \times_{I_D} I_C$$

*is an isomorphism. Moreover,*

$$(11) \quad 0 \longrightarrow I_A \longrightarrow I_B \oplus I_C \longrightarrow I_D \longrightarrow 0$$

*is strict exact.*

*Proof.* We use the controlled lifts supplied in the proof of lemma 5.1. Let  $(x_B, x_C) \in I_B \times_{I_D} I_C$ , and choose lifts  $u_B \in T_B^m$  and  $u_C \in T_C^m$  of  $x_B$  and  $x_C$  whose norms are bounded by fixed multiples of  $\|x_B\|$  and  $\|x_C\|$ . In  $T_D^m$  the difference

$$w = (u_B)_D - (u_C)_D$$

belongs to  $\ker(d_{1,D})$ . Choose a controlled lift  $v_D \in \bigwedge^2 T_D^m$  with  $d_{2,D}(v_D) = w$ , and lift  $v_D$  coefficientwise to  $v_C \in \bigwedge^2 T_C^m$ . The coefficientwise section  $T_C \rightarrow T_D$  preserves the norm. Replacing  $u_C$  by

$$u'_C = u_C + d_{2,C}(v_C)$$

does not change  $d_{1,C}(u_C)$ , and the images of  $u_B$  and  $u'_C$  in  $T_D^m$  now agree. Exactness of (10), applied coefficientwise, gives  $u_A \in T_A^m$  mapping to this pair. Then  $d_{1,A}(u_A)$  maps to  $(x_B, x_C)$ .

The same construction keeps track of the norms. The controlled lifts for  $d_{1,B}$ ,  $d_{1,C}$  and  $d_{2,D}$ , continuity of  $d_{2,C}$ , and the norm-one coefficientwise section give a constant  $C$ , independent of  $(x_B, x_C)$ , for which the lift  $x_A = d_{1,A}(u_A)$  may be chosen with

$$\|x_A\| \leq C \max\{\|x_B\|, \|x_C\|\}.$$

The map  $T_A \rightarrow T_B \oplus T_C$  is injective, so this proves both the algebraic pullback assertion and the boundedness of its inverse.

It remains to see that  $I_A$  is closed. The ideals  $I_B$  and  $I_C$  are closed by lemma 5.1. If  $x$  lies in the closure of  $I_A$ , its images lie in  $I_B$  and  $I_C$ . The controlled pullback just proved gives  $x_A \in I_A$  with the same two images as  $x$ . Injectivity of  $T_A \rightarrow T_B \oplus T_C$  gives  $x = x_A$ , as required.

Finally, let  $y \in I_D$ . A controlled lift of  $y$  through  $d_{1,D}$ , followed by the coefficientwise section  $T_D^m \rightarrow T_C^m$ , produces  $x_C \in I_C$  mapping to  $y$ , with norm bounded by a fixed multiple of  $\|y\|$ . Thus  $I_B \oplus I_C \rightarrow I_D$  is a strict surjection. Its kernel is  $I_A$  by the pullback assertion, and the preceding norm estimate makes the inclusion on the left strict. This proves (11).  $\square$

**Proposition 5.3.** *For every rational datum  $\alpha$  in  $A$ , the canonical sequence*

$$(12) \quad 0 \longrightarrow A_\alpha \longrightarrow B_\alpha \oplus C_\alpha \longrightarrow D_\alpha \longrightarrow 0$$

*is strict exact, and  $C_\alpha \rightarrow D_\alpha$  is a strict surjection. Equivalently,*

$$A_\alpha \cong B_\alpha \times_{D_\alpha} C_\alpha.$$

*The coefficient maps used in these identifications commute with rational restriction.*

*Proof.* The ideals  $I_B, I_C, I_D$  are closed by lemma 5.1, and  $I_A$  is closed by lemma 5.2. Hence (3) gives

$$E_\alpha = T_E/I_E \quad (E = A, B, C, D).$$

We first prove algebraic exactness. If a representative  $p \in T_A$  maps into both  $I_B$  and  $I_C$ , the controlled pullback in lemma 5.2 gives an element of  $I_A$  with the same two images. Injectivity of  $T_A \rightarrow T_B \oplus T_C$  then shows that  $p \in I_A$ . This gives injectivity on the left. Conversely, take compatible classes in  $T_B/I_B$  and  $T_C/I_C$  and choose representatives  $b$  and  $c$ . Their defect in  $T_D$  belongs to  $I_D$ . Lift this defect to an element of  $I_C$  using the strict surjection in (11), and add it to  $c$ . The corrected representatives agree in  $T_D$ , so (10) lifts them to  $T_A$ . This proves (12).

For the topology on the left, let  $x \in A_\alpha$  and put

$$M = \max\{\|x_B\|, \|x_C\|\}.$$

Choose representatives  $b \in T_B$  and  $c \in T_C$  whose norms are arbitrarily close to the quotient norms of  $x_B$  and  $x_C$ . Their defect lies in  $I_D$ . The controlled surjection  $I_C \rightarrow I_D$  gives a correction  $y \in I_C$  whose norm is bounded by a fixed multiple of the norm of this defect. Thus



$b$  and  $c + y$  agree in  $T_D$  and lift to  $p \in T_A$ . The max-norm identity for the ambient pullback gives, after letting the error in the representatives tend to zero,

$$\|x\| \leq CM$$

for a constant  $C$  depending only on the datum. The two quotient maps are norm-nonincreasing, so this estimate says precisely that  $A_\alpha \rightarrow B_\alpha \oplus C_\alpha$  is a topological embedding.

The coefficientwise section  $T_D \rightarrow T_C$  shows directly that  $C_\alpha \rightarrow D_\alpha$  is surjective. It also shows that  $T_C \rightarrow T_D$  is open; passing to quotient seminorms proves that the induced map is open. Together with continuity, this makes  $C_\alpha \rightarrow D_\alpha$  a strict surjection.

All maps used above come from the coefficient maps  $A \rightarrow B, C$  and  $B, C \rightarrow D$ . More generally, a homomorphism  $E \rightarrow E'$  carrying a rational datum  $\alpha$  to  $\alpha'$  induces a coefficientwise map  $T_E \rightarrow T_{E'}$  which sends the relations for  $\alpha$  to those for  $\alpha'$ . The pullback identifications therefore commute with these maps. Applying this to rational restriction, and then using presentation independence and transitivity, proves the asserted compatibility under refinement.  $\square$

## 6. SHEAFINESS

**Theorem 6.1.** *The Huber pair  $(A, A^\circ)$  is sheafy.*

*Proof.* Give  $B, C, D$  their maximal rings of integral elements. The max/Gauss estimates (7) show that every structural map is norm-nonincreasing. Consequently it sends bounded subsets to bounded subsets; in particular, if  $x$  is power-bounded, then the set of powers of its image is the image of the bounded set  $\{x^n : n \geq 0\}$ . The structural maps therefore send power-bounded elements to power-bounded elements and define morphisms of Huber pairs.

Each of  $B, C, D$  is an affinoid  $k$ -algebra and is therefore sheafy by Huber's theorem [7, Theorem 2.2]. Let  $U = \bigcup_i U_i$  in  $\mathrm{Spa}(A, A^\circ)$ . Push the datum for  $U$  and the data for the  $U_i$  to  $B, C, D$ . If a section over  $U$  restricts to zero on every  $U_i$ , its images over  $B$  and  $C$  vanish by separatedness there. Injectivity of the localised Milnor row from proposition 5.3 then shows that the original section is zero.

Now take a matching family over the  $U_i$ . Its images glue over  $B$  and  $C$ . The two glued sections have the same image over  $D$ , since their restrictions agree on the pushed cover and the presheaf for  $D$  is separated. Exactness of the localised Milnor row over  $U$  gives a section over  $A$ , and injectivity of the corresponding rows over the  $U_i$  shows that it restricts to the given family.

It remains to retain the topology. The image of the product restriction map is the matching-family equaliser just described, which is closed in the finite product of the section rings. The restriction map is continuous and injective, so the nonarchimedean closed-range theorem, using the topologically nilpotent unit of  $A$ , makes it a topological embedding. This proves the sheaf condition on finite rational covers. The sheaf-on-a-rational-basis theorem [15, Tag 009O] then gives the all-open topological sheaf condition for  $(A, A^\circ)$ .  $\square$

*Proof of theorem 1.1.* Completeness, uniformity, nonnoetherianity, and the domain property are proposition 3.1; sheafiness is theorem 6.1; and failure of stable uniformity is proposition 4.2.  $\square$

## 7. A SECOND EXAMPLE

The finite-jet construction above loses uniformity because separated completion creates a nilpotent. We now give a second construction in which the bad rational localisation is an integral domain. The loss of uniformity is instead caused by an unbounded family of nonnilpotent power-bounded elements.

**Theorem 7.1.** *There is a complete Tate  $k$ -algebra  $A$  with the following properties.*

- (1)  $\mathcal{A}$  is a uniform, nonnoetherian integral domain and  $\mathcal{A}^\circ = \mathcal{A}_0$  for the ring of definition displayed below.
- (2) The Huber pair  $(\mathcal{A}, \mathcal{A}^\circ)$  is sheafy.
- (3) The rational localisation

$$\mathcal{B} = \mathcal{A} \left\langle \frac{W}{\varpi} \right\rangle$$

is an integral domain but is not uniform.

Consequently  $\mathcal{A}$  is not stably uniform. In particular, failure of stable uniformity need not be caused by a nilpotent in the bad rational localisation.

**7.1. The weighted-parity algebra.** Let  $I = \mathbf{N}_{>0}$ . For a finite-support multi-index  $\nu = (\nu_n)_{n \in I} \in \mathbf{N}^{(I)}$ , put

$$(13) \quad \omega(\nu) = \sum_{\substack{n \geq 1 \\ \nu_n \text{ odd}}} n.$$

Consider the submonoid

$$(14) \quad S = \left\{ (a, \nu) \in \mathbf{N} \times \mathbf{N}^{(I)} : a \geq \omega(\nu) \right\}.$$

The inequality

$$(15) \quad \omega(\nu + \mu) \leq \omega(\nu) + \omega(\mu)$$

shows that  $S$  is closed under addition. It is locally finite: a fixed element of  $S$  has only finitely many decompositions as a sum of two elements of  $S$ .

Let  $k[S]$  be the monoid algebra with basis  $W^a U^\nu$ , where  $U^\nu = \prod_n U_n^{\nu_n}$ . Give it the Gauss norm

$$(16) \quad \left\| \sum_{(a, \nu) \in S} c_{a, \nu} W^a U^\nu \right\|_G = \sup_{a, \nu} |c_{a, \nu}|.$$

Its completion is

$$(17) \quad \mathcal{A} = \left\{ \sum_{(a, \nu) \in S} c_{a, \nu} W^a U^\nu : c_{a, \nu} \in k, \quad c_{a, \nu} \longrightarrow 0 \right\},$$

where convergence to zero is over the discrete index set  $S$ . Local finiteness makes the coefficientwise convolution finite, and the submultiplicativity of the Gauss norm on  $k[S]$  extends multiplication continuously to  $\mathcal{A}$ . Put

$$(18) \quad \mathcal{A}_0 = \left\{ \sum_{(a, \nu) \in S} c_{a, \nu} W^a U^\nu \in \mathcal{A} : c_{a, \nu} \in k^\circ \right\}.$$

Then  $\mathcal{A}_0$  is the closed unit ball for (16), it is  $\varpi$ -adically complete, and  $\mathcal{A} = \mathcal{A}_0[1/\varpi]$ .

For  $N \geq 1$ , let  $\mathcal{A}_N \subset \mathcal{A}$  be the closed subring supported on monomials involving only  $W, U_1, \dots, U_N$ . Put

$$Y_n = W^n U_n, \quad Z_n = U_n^2,$$

and let  $\mathcal{T}_N \subset \mathcal{A}_N$  be the subring supported on even  $U_n$ -exponents.

**Lemma 7.2** (Finite heads). *The substitution  $Z_n \mapsto U_n^2$  gives a norm-preserving ring isomorphism*

$$k\langle W, Z_1, \dots, Z_N \rangle \xrightarrow{\sim} \mathcal{T}_N.$$

*Every  $f \in \mathcal{A}_N$  has a parity-slice expansion*

$$(19) \quad f = \sum_{\epsilon \in \{0,1\}^N} f_\epsilon Y^\epsilon, \quad f_\epsilon \in \mathcal{T}_N.$$

*Consequently  $\mathcal{A}_N$  is finite over  $\mathcal{T}_N$  and is a strongly noetherian Tate ring. The inclusions between the heads and into  $\mathcal{A}$  preserve the norm. Moreover, for every  $f \in \mathcal{A}$  and  $\ell \geq 0$  there are  $N$  and  $f_N \in \mathcal{A}_N$  such that*

$$\|f - f_N\|_G \leq |\varpi|^\ell \|f\|_G.$$

*In particular,*

$$(20) \quad \mathcal{A} = \widehat{\bigcup_{N \geq 1} \mathcal{A}_N}.$$

*Proof.* On even exponents, halving the  $U_n$ -coordinates identifies the support monoid with that of the Tate algebra  $k\langle W, Z_1, \dots, Z_N \rangle$ ; the coefficient map and its inverse preserve the Gauss norm. Splitting each  $U_n$ -exponent according to its parity gives (19). The finitely many  $Y^\epsilon$  therefore generate  $\mathcal{A}_N$  over  $\mathcal{T}_N$ . The same argument after adjoining any finite number of Tate variables gives strong noetherianity. The inclusions are coefficientwise and hence isometric. Finally, truncating a restricted series to a sufficiently large head gives the stated estimate and hence (20).  $\square$

**Proposition 7.3.** *The ring  $\mathcal{A}$  is a complete uniform Tate  $k$ -algebra and an integral domain, with*

$$\mathcal{A}^\circ = \mathcal{A}_0.$$

*Proof.* The Gauss norm on the countable restricted Tate algebra  $k\langle W, U_1, U_2, \dots \rangle$  is multiplicative. To see this, note first that the norm of a nonzero restricted series is attained. For each of two such series choose, among the norm-attaining exponents, the lexicographically largest one. In the coefficient of the sum of these two exponents, the product of the chosen coefficients strictly dominates every other convolution term. The ultrametric inequality therefore gives

$$\|fg\|_G = \|f\|_G \|g\|_G.$$

This is the countable restricted-series argument used in the formalisation, whose underlying Lean implementation is adapted from work of William Coram [5]. The support algebra  $\mathcal{A}$  is a closed isometric subring of this Tate algebra, so its Gauss norm is multiplicative and it is a domain.

If  $x \in \mathcal{A}_0$ , all powers of  $x$  remain in  $\mathcal{A}_0$ . If  $x \notin \mathcal{A}_0$ , multiplicativity gives  $\|x^m\|_G = \|x\|_G^m$ , which is unbounded. Hence  $\mathcal{A}^\circ = \mathcal{A}_0$ . Completeness and the Tate property follow from (17) and the topologically nilpotent unit  $\varpi$ .  $\square$

**Proposition 7.4.** *The ring  $\mathcal{A}$  is not noetherian.*

*Proof.* For  $m \geq 1$ , let

$$I_m = (Z_1, \dots, Z_m) \subset \mathcal{A}.$$

Define a norm-nonincreasing homomorphism

$$\psi_m : \mathcal{A} \longrightarrow k\langle T \rangle,$$

by retaining only the coefficients of the monomials  $U_{m+1}^{2q}$  and sending that monomial to  $T^q$ . The support condition shows directly that this is multiplicative: a product can contribute such a monomial only when both factors contribute monomials of the same form. Thus  $\psi_m$  sends  $W$  and every  $Y_j$  to zero, sends  $Z_{m+1}$  to  $T$ , and sends every other  $Z_j$  to zero. It kills  $I_m$  but not  $Z_{m+1}$ . Therefore

$$I_1 \subsetneq I_2 \subsetneq I_3 \subsetneq \cdots,$$

so  $\mathcal{A}$  is not noetherian.  $\square$

**7.2. A reduced nonuniform rational chart.** The ideal  $(W, \varpi)$  is open because  $\varpi$  is a unit of  $\mathcal{A}$ . Define the complete  $k^\circ$ -algebra

$$(21) \quad \mathcal{B}_0 = \left\{ \sum_{\nu \in \mathbf{N}^{(I)}, d \geq \omega(\nu)} b_{d,\nu} X^d U^\nu : b_{d,\nu} \in \varpi^{\omega(\nu)} k^\circ, \quad \varpi^{-\omega(\nu)} b_{d,\nu} \rightarrow 0 \right\},$$

with norm

$$(22) \quad \left\| \sum b_{d,\nu} X^d U^\nu \right\|_{\text{in}} = \sup_{d,\nu} |\varpi|^{-\omega(\nu)} |b_{d,\nu}|.$$

The subadditivity (15) shows that  $\mathcal{B}_0$  is a ring. Put  $\mathcal{B} = \mathcal{B}_0[1/\varpi]$ .

**Proposition 7.5.** *There is a bicontinuous ring isomorphism*

$$(23) \quad \mathcal{A} \left\langle \frac{W}{\varpi} \right\rangle \xrightarrow{\sim} \mathcal{B}, \quad W \mapsto \varpi X.$$

*Proof.* The datum  $(W; \varpi)$  already lies in the zeroth head  $H_0 = k\langle W \rangle$ . Let

$$Q = H_0 \langle T \rangle / \overline{\langle \varpi T - W \rangle}$$

be its separated completed quotient. Evaluation at  $W = \varpi X$  and  $T = X$  gives a continuous map  $Q \rightarrow k\langle X \rangle$ . Conversely, evaluation of  $k\langle X \rangle$  at the image of  $T$  gives a continuous map in the other direction. The two maps are inverse by polynomial density, and the norm estimates in the two universal properties show that the isomorphism is isometric.

Apply the coefficientwise-localisation construction of proposition 7.8 to this zeroth head. It identifies the rational chart bicontinuously with the  $c_0$  tail algebra over  $Q$ . After replacing  $Q$  by  $k\langle X \rangle$ , this is the model (21); its sup norm is (22), since the tail basis element of parity  $\nu$  contributes the factor  $\varpi^{\omega(\nu)}$ . The head evaluation sends  $W$  to  $\varpi X$ .  $\square$

**Proposition 7.6.** *The ring  $\mathcal{B}$  is an integral domain but is not uniform.*

*Proof.* Group a series in  $\mathcal{B}$  by its  $U$ -multi-index. This gives a coefficientwise homomorphism

$$\mathcal{B} \longrightarrow k\langle X \rangle[[U_1, U_2, \dots]].$$

It is injective, and the target is a domain because  $k\langle X \rangle$  is a domain. Hence  $\mathcal{B}$  is a domain. In the finite-head formalisation below, this is the embedding of the corresponding  $c_0$  tail algebra into formal multivariable power series; the twisting element is  $\varpi X$ , which is a non-zero-divisor.

Inside  $\mathcal{A}$  write  $Y_n = W^n U_n$  and put

$$T_n = \varpi^{-n} Y_n = X^n U_n \in \mathcal{B}.$$

Formula (22) gives

$$\|T_n\|_{\text{in}} = |\varpi|^{-n}, \quad \|T_n^2\|_{\text{in}} = 1.$$

It follows by separating even and odd powers that the powers of each fixed  $T_n$  are bounded. Thus every  $T_n$  is power-bounded. The family  $(T_n)_{n \geq 1}$  is not bounded, since its norms  $|\varpi|^{-n}$  are unbounded. Hence  $\mathcal{B}^\circ$  is unbounded and  $\mathcal{B}$  is not uniform.  $\square$

**7.3. Small perturbations of rational data.** The finite-head argument rests on the following elementary stability result.

**Lemma 7.7** (Small perturbation lemma). *Let  $E$  be a complete nonarchimedean normed commutative ring, let  $E^+ \subset E$  be a plus ring containing the closed unit ball  $E_0 = \{x : \|x\| \leq 1\}$ , and let  $t \in E^\times$  satisfy*

$$0 < \|t\| < 1, \quad \|tx\| = \|t\|\|x\| \quad (x \in E).$$

*Let  $\alpha = (f_1, \dots, f_m; g)$  be a rational datum with entries in  $E_0$ . Write  $d_0 = g$  and  $d_i = f_i$  for  $i \geq 1$ . Suppose that for some  $\ell \geq 0$  there are  $a_0, \dots, a_m \in E_0$  with*

$$(24) \quad t^\ell = \sum_{j=0}^m a_j d_j.$$

*Let  $d'_j \in E_0$  satisfy*

$$(25) \quad \|d'_j - d_j\| \leq \|t\|^{\ell+1} \quad (0 \leq j \leq m),$$

*and write  $\alpha' = (f'_1, \dots, f'_m; g')$ . Then  $\alpha'$  is a rational datum,  $\alpha$  and  $\alpha'$  define the same rational subset of  $\text{Spa}(E, E^+)$ , and their completed rational localisations are bicontinuously isomorphic by an isomorphism commuting with the canonical maps from  $E$ .*

*Proof.* The scaling hypothesis and the fact that  $t$  is a unit give  $d'_j - d_j = t^{\ell+1} h_j$  with  $\|h_j\| \leq 1$ . Hence

$$\sum_j a_j d'_j = t^\ell (1 + tH), \quad H = \sum_j a_j h_j, \quad \|H\| \leq 1.$$

Since  $\|tH\| < 1$ , the element  $1 + tH$  is a unit with inverse in  $E_0$ . Multiplying the  $a_j$  by this inverse gives a unit-ball Bezout relation for the primed datum; in particular,  $\alpha'$  is rational.

Let  $x$  lie in the rational subset defined by  $\alpha$ . Since  $a_j \in E_0 \subseteq E^+$ , relation (24) gives

$$|t(x)|^\ell \leq |g(x)|.$$

The perturbations have value at most  $|t(x)|^{\ell+1} < |g(x)|$ , so  $|g'(x)| = |g(x)|$  and  $|f'_i(x)| \leq |g'(x)|$ . This proves one inclusion of rational subsets. Applying the same argument to the primed Bezout relation proves the reverse inclusion.

Rational localisation is independent of the datum used to present a rational subset. The equality of the two subsets therefore gives restriction maps in both directions between  $E_\alpha$  and  $E_{\alpha'}$ . Transitivity of rational restriction shows that their composites are the identity. The two restriction maps are continuous and commute with the canonical maps from  $E$ , as required.  $\square$

**7.4. Finite-head rational localisation.** Fix  $N \geq 1$ . For a tail multi-index  $\mu = (\mu_n)_{n>N} \in \mathbf{N}^{\{n>N\}}$ , write  $\mu_n = 2q_n + \epsilon_n$  with  $\epsilon_n \in \{0, 1\}$  and put

$$(26) \quad e_\mu = \prod_{n>N} Y_n^{\epsilon_n} Z_n^{q_n}.$$

The finite-stage normal forms give an isometric Banach-module decomposition

$$(27) \quad \mathcal{A} \cong \widehat{\bigoplus_\mu}^{c_0} \mathcal{A}_N e_\mu.$$

Multiplication is determined by

$$(28) \quad e_\mu e_\lambda = W^{\omega(\mu) + \omega(\lambda) - \omega(\mu + \lambda)} e_{\mu + \lambda}.$$

Here  $\omega$  is applied only to the tail coordinates. Projection to the coefficient of  $e_0$  is a norm-nonincreasing algebra retraction

$$(29) \quad \rho_N : \mathcal{A} \longrightarrow \mathcal{A}_N.$$

**Proposition 7.8** (Coefficientwise localisation). *Let  $\alpha$  be a rational datum in  $\mathcal{A}_N$ , and put  $P = (\mathcal{A}_N)_\alpha$ . There is a bicontinuous ring isomorphism*

$$(30) \quad \mathcal{A}_\alpha \xrightarrow{\sim} \widehat{\bigoplus_\mu^{c_0} P e_\mu}.$$

*It is natural for rational refinements represented in a finite head: if  $P \rightarrow Q$  is the corresponding head restriction map, the induced restriction sends*

$$(31) \quad \sum_\mu p_\mu e_\mu \longmapsto \sum_\mu (p_\mu|_Q) e_\mu.$$

*The same statement holds after increasing  $N$ .*

*Proof.* Let

$$R = \mathcal{A}_N \langle T_1, \dots, T_m \rangle, \quad J = \overline{(gT_i - f_i)_i} \subset R, \quad Q = R/J.$$

Evaluation at  $T_i = f_i/g$  identifies  $Q$  with the head rational localisation  $P$ .

The tail decomposition applies after adjoining the variables  $T_i$ . Apply the quotient map  $R \rightarrow Q$  to each tail coefficient. Since the rational relations lie in the head, this gives a continuous homomorphism

$$\Phi : \mathcal{A}_\alpha \longrightarrow \widehat{\bigoplus_\mu^{c_0} Q e_\mu}.$$

Conversely, the canonical map from  $Q$  to  $\mathcal{A}_\alpha$  may be applied to every coefficient. For a null family  $(q_\mu)_\mu$  the resulting series

$$\sum_\mu q_\mu e_\mu$$

converges in  $\mathcal{A}_\alpha$ , and defines a continuous homomorphism  $\Psi$  in the other direction. The multiplication rule (28) shows that both maps are ring homomorphisms.

The two composites are the identity on finite tail sums with polynomial head coefficients. These are dense on both sides, so continuity gives  $\Psi\Phi = 1$  and  $\Phi\Psi = 1$ . Composing with  $Q \cong P$  proves (30).

For a rational refinement in the head, the two evaluation maps commute with the head restriction map. The preceding construction is therefore coefficientwise natural and gives (31). Presentation independence and transitivity give the same conclusion after enlarging the head.  $\square$

**Corollary 7.9** (Finite-head presentation). *Every rational localisation of  $\mathcal{A}$  is bicontinuously isomorphic to a ring of the form*

$$(32) \quad \widehat{\bigoplus_\mu^{c_0} P e_\mu},$$

*where  $P$  is a rational localisation of some strongly noetherian Tate head  $\mathcal{A}_N$ .*

*Proof.* Start with a rational datum in  $\mathcal{A}$ . Multiplying all entries by one power of  $\varpi$  does not change the rational subset or localisation, so assume that the entries lie in  $\mathcal{A}_0$ . Choose an integral Bezout relation

$$\varpi^\ell = \sum_j a_j d_j, \quad a_j \in \mathcal{A}_0.$$

By (20), for some  $N$  the entries  $d_j$  admit approximations  $d'_j \in \mathcal{A}_N \cap \mathcal{A}_0$  satisfying  $\|d'_j - d_j\| \leq |\varpi|^{\ell+1}$ . By lemma 7.7, the primed datum gives the same rational subset and the same completed localisation. The primed integral Bezout relation produced in that lemma remains valid after applying the retraction  $\rho_N$ ; hence the primed datum is a rational datum already in  $\mathcal{A}_N$ . Now apply proposition 7.8.  $\square$

### 7.5. Sheafiness.

**Theorem 7.10.** *The Huber pair  $(\mathcal{A}, \mathcal{A}^\circ)$  is sheafy.*

*Proof.* We first work with a finite rational cover  $U = \bigcup_i U_i$ . Approximate the datum for  $U$  and the data for all the  $U_i$  simultaneously in one sufficiently large finite head. The small perturbation lemma does not change the rational domains, and proposition 7.8 gives compatible identifications

$$\mathcal{O}_X(U) \cong \widehat{\bigoplus_\mu}^{c_0} P e_\mu, \quad \mathcal{O}_X(U_i) \cong \widehat{\bigoplus_\mu}^{c_0} P_i e_\mu,$$

with coefficientwise restriction maps. Here  $P$  is a rational localisation of one strongly noetherian Tate head and the  $P_i$  are rational localisations of  $P$ . The domains defined by the  $P_i$  cover  $\mathrm{Spa}(P, P^\circ)$ : pull a point back along the split coefficient retraction to the tail algebra, use the original cover, and then apply the retraction again.

Let  $(x_i)_i$  be a matching family and write  $x_i = \sum_\mu x_{i,\mu} e_\mu$ . Since  $P$  is sheafy, the matching family  $(x_{i,\mu})_i$  glues to a unique  $q_\mu \in P$  for every  $\mu$ . It remains to check that  $(q_\mu)_\mu$  is a null family. Its restrictions to every  $P_i$  tend to zero because each  $x_i$  belongs to a  $c_0$ -sum. The product restriction map

$$P \longrightarrow \prod_i P_i$$

is a topological embedding, so it reflects convergence to zero. Hence  $q_\mu \rightarrow 0$ , and  $\sum_\mu q_\mu e_\mu$  is the required section over  $U$ . The same coefficientwise argument proves separation.

The image of the product restriction map for  $\mathcal{A}$  is therefore the matching-family equaliser. This equaliser is closed, and the restriction map is continuous and injective. The nonarchimedean closed-range theorem over the Tate ring  $\mathcal{A}$  now makes it a topological embedding. We have proved the topological sheaf condition on finite rational covers. The general rational-basis theorem upgrades this to the all-open structure presheaf, and its plus-ring independence gives the same conclusion for every ring of integral elements. In particular,  $(\mathcal{A}, \mathcal{A}^\circ)$  is sheafy.  $\square$

*Proof of theorem 7.1.* The uniform domain assertion is proposition 7.3, and nonnoetherianity is proposition 7.4. Sheafiness is theorem 7.10. Finally, propositions 7.5 and 7.6 identify the rational chart  $|W| \leq |\varpi|$  as a nonuniform integral domain.  $\square$

**Remark 7.11** (Relation with the first construction). The finite-jet and weighted-parity examples use different mechanisms. In the finite-jet ring, a whole tail becomes infinitely  $\varpi$ -divisible and the bad chart acquires a nilpotent. In the weighted-parity ring, every rational calculation can be moved into a noetherian finite head and then carried coefficientwise through a  $c_0$ -tail. Its bad chart remains a domain, and nonuniformity is witnessed by the unbounded family  $(T_n)_{n \geq 1}$  of nonnilpotent power-bounded elements. The use of an infinite monomial support and of unbounded power-bounded directions is in the tradition of the examples of Buzzard–Verberkmoes and Mihara; the finite-head coefficientwise descent is the additional feature that makes sheafiness accessible here.

## APPENDIX A. LEAN FORMALISATION AND VERIFICATION

This appendix records the Lean formalisation of the two examples. The finite-jet declarations are taken from AINTLIB commit `b007a4f3d`, and the weighted-parity declarations from commit `090a28921` [1]. For each example Lean proves the Tate structure, uniformity, domain and nonnoetherianity statements, sheafiness, and failure of stable uniformity, together with the rational chart calculation used in the proof.

Neither pinned snapshot was available from the public AINTLIB repository at the time of writing. The interactive version of the paper includes each declaration used below, extracted from the appropriate commit. The verification report records the Lean and mathlib versions, the build, and the axiom checks.

**A.1. Sheafiness.** The finite-jet proof first establishes a condition on finite rational covers. Its historical internal class is `IsSheafy`, in the `ValuationSpectrum` namespace and defined in `StructureSheaf.lean`.

```
class IsSheafy (A : Type u) [CommRing A] [TopologicalSpace A]
  [IsTopologicalRing A] [inst₁ : PlusSubring A]
  [inst₂ : IsHuberRing A] [T2Space A] [NonarchimedeanRing A]
  [letI : UniformSpace A :=
    IsTopologicalAddGroup.rightUniformSpace A; CompleteSpace A]
  [IsRingOfIntegralElements (A+)] : Prop where
embedding : ∀ (C : RationalCoveringData A), C.IsRational →
  Topology.IsEmbedding (productRestrictionSub A C)
gluing : ∀ (C : RationalCoveringData A), C.IsRational →
  ∀ (f : ∀ (D : ↑C.covers), presheafValue D.1),
  (∀ (D₁ D₂ : ↑C.covers) (D₃ : RationalLocData A)
    (h₃₁ : rationalOpen D₃.T D₃.s ⊆
      rationalOpen D₁.1.T D₁.1.s)
    (h₃₂ : rationalOpen D₃.T D₃.s ⊆
      rationalOpen D₂.1.T D₂.1.s),
    restrictionMap D₁.1 D₃ h₃₁ (f D₁) =
      restrictionMap D₂.1 D₃ h₃₂ (f D₂)) →
  ∃ x : presheafValue C.base, ∀ (D : ↑C.covers),
    restrictionMap C.base D.1 (C.hsubset D.1 D.2) x = f D
```

The two fields say that restriction to a finite rational cover is a topological embedding and that every matching family glues. Compatibility is written using common rational refinements; the exact-intersection theorem identifies this with the usual compatibility on pairwise intersections. Thus `IsSheafy A` is the topological sheaf condition on the rational basis.

For an arbitrary open subset  $V \subseteq \text{Spa}(A, A^+)$ , `limitSections V` is the projective limit of the sections over the rational subdomains contained in  $V$ . The corresponding all-open condition is `IsLimitSheaf`, from `SheafyPair.lean`.

```
structure IsLimitSheaf (A : Type u) [CommRing A] [TopologicalSpace A]
  [IsTopologicalRing A] [PlusSubring A] [IsHuberRing A]
  [HasLocLiftPowerBounded A] : Prop where
injective : ∀ {V : Opens ↑(Spa A A+)} {ι : Type u}
  {U : ι → Opens ↑(Spa A A+)}
  (hle : ∀ i, U i ≤ V)
  (⊆ : (V : Set ↑(Spa A A+)) ⊆ ⋃ i, (U i : Set ↑(Spa A A+)))
  {x y : ↑(limitSections V)},
  (∀ i, limitRestrict (hle i) x = limitRestrict (hle i) y) → x = y
glue : ∀ {V : Opens ↑(Spa A A+)} {ι : Type u}
```



```

{U :  $\iota \rightarrow \text{Opens } \uparrow(\text{Spa } A \ A^+)$ }
(hle :  $\forall i, U \ i \leq V$ )
( $\_ : (V : \text{Set } \uparrow(\text{Spa } A \ A^+)) \subseteq \bigcup i, (U \ i : \text{Set } \uparrow(\text{Spa } A \ A^+))$ )
(s :  $\forall i, \uparrow(\text{limitSections } (U \ i))$ ),
( $\forall i \ j$ ,
  limitRestrict (inf_le_left (a := U i) (b := U j)) (s i) =
  limitRestrict (inf_le_right (a := U i) (b := U j)) (s j))  $\rightarrow$ 
 $\exists x : \uparrow(\text{limitSections } V), \forall i, \text{limitRestrict } (hle \ i) \ x = s \ i$ 
isEmbedding :  $\forall \{V : \text{Opens } \uparrow(\text{Spa } A \ A^+)\} \{ \iota : \text{Type } u \}$ 
{U :  $\iota \rightarrow \text{Opens } \uparrow(\text{Spa } A \ A^+)$ }
(hle :  $\forall i, U \ i \leq V$ )
( $\_ : (V : \text{Set } \uparrow(\text{Spa } A \ A^+)) \subseteq \bigcup i, (U \ i : \text{Set } \uparrow(\text{Spa } A \ A^+))$ ),
Topology.IsEmbedding (limitRestrictProd hle)

```

The fields `injective` and `glue` are the usual sheaf axioms on arbitrary opens. The final field says that the resulting map to the matching-family equaliser is also a topological embedding. This is the extra condition needed for a sheaf of complete topological rings.

For complete Tate pairs the rational-basis and all-open predicates are equivalent:

```

theorem isSheafy_iff_isLimitSheaf :
  IsSheafy A  $\leftrightarrow$  IsLimitSheaf A :=
  ⟨fun _ => isLimitSheaf_of_isSheafy,
   isSheafy_of_isLimitSheaf⟩

```

The forward implication is the sheaf-on-a-rational-basis theorem, and the converse is restriction to rational covers. The structure-presheaf equivalence identifies `IsLimitSheaf A` with Wedhorn’s topological-ring formulation [16, Remark 8.20]. Mathlib’s ordinary categorical sheaf predicate is weaker because it does not record the topology.

For a complete Tate ring the following definitions fix a ring of integral elements and then remove the choices of plus ring and completion. They are in `SheafyRing.lean`.

```

def IsSheafyFor (Aplus : RingOfIntegralElements A) : Prop :=
  let I := Aplus.toPlusSubring
  have I : IsRingOfIntegralElements (A+ : Subring A) := Aplus.2
  have I : HasLocLiftPowerBounded A := hasLocLiftPowerBounded_faithful
  IsLimitSheaf A

def IsSheafyComplete : Prop :=
   $\forall$  Aplus : RingOfIntegralElements A, IsSheafyFor A Aplus

def IsSheafyTateRing : Prop :=
   $\forall$  (P : PairOfDefinition A)
    (Bplus : RingOfIntegralElements (CompletionModel A P)),
    IsSheafyFor (CompletionModel A P) Bplus

```

The predicate `IsSheafyFor A Aplus` says that the Huber pair  $(A, A^+)$  is sheafy. For complete Tate rings, the chosen-plus-ring equivalence shows that this is independent of the chosen ring of integral elements. The equivalences in `SheafyCompletionModel.lean` remove the choice of completion. Consequently `IsSheafyTateRing` is the analytic specialisation of Wedhorn’s Definition 8.26 [16, Definition 8.26].

**A.2. Strong noetherianity.** The definitions below are from `RestrictedPowerSeries.lean`.

```

def MvPowerSeries.IsRestricted {k :  $\mathbb{N}$ } {A : Type*}
  [CommRing A] [TopologicalSpace A]

```

```

(f : MvPowerSeries (Fin k) A) : Prop :=
Tendsto (fun s : Fin k →0 ℕ => MvPowerSeries.coeff s f)
cofinite (nhds 0)

class IsStronglyNoetherian (A : Type*) [CommRing A]
[TopologicalSpace A] [NonarchimedeanRing A] : Prop where
isNoetherianRing_restricted : ∀ k : ℕ,
IsNoetherianRing (restrictedMvPowerSeriesSubring k A)

```

The cofinite convergence condition is the usual condition that the coefficients tend to zero, so `restrictedMvPowerSeriesSubring k A` represents  $A\langle T_1, \dots, T_k \rangle$ . The second declaration therefore says that these rings are noetherian for every  $k$ . Since  $A$  is already complete, this is Wedhorn’s definition of strong noetherianity [16, Proposition and Definition 6.36].

**A.3. The finite-jet example.** We finally record the definition of the example itself. In the following code, `PowerSeries.Restricted R 1` is the radius-one Tate algebra in one variable over  $R$ , `RestrictedLaurent K` is  $K\langle W, W^{-1} \rangle$ , and `DualNumber R` is  $R[Q]/(Q^2)$ . The source is `FJP/Over/JetRings.lean`.

```

abbrev L : Type u := RestrictedLaurent K

abbrev JetC : Type u := PowerSeries.Restricted (L K) (1 : ℝ)

abbrev JetB : Type u :=
DualNumber (PowerSeries.Restricted K (1 : ℝ))

abbrev JetD : Type u := DualNumber (L K)

noncomputable def jetSupport : Subring (JetC K) where
carrier := {f |
  qCoeff K 0 f ∈ nonnegSubring K ∧
  qCoeff K 1 f ∈ nonnegSubring K}
zero_mem' := by
  constructor <=> rw [qCoeff_zero] <=> exact zero_mem _
one_mem' := by
  constructor <=> rw [qCoeff_one]
  · rw [if_pos rfl]; exact one_mem _
  · rw [if_neg one_ne_zero]; exact zero_mem _
add_mem' := fun {f g} hf hg => by
  constructor <=> rw [qCoeff_add]
  · exact add_mem hf.1 hg.1
  · exact add_mem hf.2 hg.2
neg_mem' := fun {f} hf => by
  constructor <=> rw [qCoeff_neg]
  · exact neg_mem hf.1
  · exact neg_mem hf.2
mul_mem' := fun {f g} hf hg => by
  constructor
  · rw [qCoeff_zero_mul]
    exact mul_mem hf.1 hg.1
  · rw [qCoeff_one_mul]
    exact add_mem (mul_mem hf.1 hg.2) (mul_mem hf.2 hg.1)

abbrev JetA : Type u := ↑(jetSupport K)

```

The predicate `nonnegSubring K` cuts out  $K\langle W \rangle$  inside  $K\langle W, W^{-1} \rangle$ . Consequently the carrier condition says that the coefficients of  $Q^0$  and  $Q^1$  have no negative powers of  $W$ , while the coefficients of  $Q^n$  for  $n \geq 2$  are unrestricted. Thus `JetA K` is

$$\left\{ f_0(W) + Qf_1(W) + Q^2h(W, W^{-1}, Q) \mid \begin{matrix} f_0, f_1 \in K\langle W \rangle \\ h \in L\langle Q \rangle \end{matrix} \right\},$$

which is the coefficient description of  $A$  in equation (5) after inverting  $\varpi$ .

The file also constructs the four maps in the Milnor square and proves `milnorRow_exact`: every compatible pair in `JetB K` and `JetC K` comes from a unique element of `JetA K`. This identifies `JetA K` with  $B \times_D C$ . The sheaf argument produces the finite-rational-cover condition displayed above:

```
include  $\varpi$  hK0 in
theorem isSheafy_JetA :
  ValuationSpectrum.IsSheafy (JetA K) where
  embedding := fun C hC =>
    productRestrictionSub_isEmbedding_JetA  $\varpi$  hK0 C hC
  gluing := fun C hC f hcompat =>
    gluing_JetA  $\varpi$  hK0 C hC f hcompat
```

Here  $\varpi$  is a `Uniformizer K`, and the remaining input is the noetherianity of `unitBall K`. This is the version of sheafiness proved directly for the example. The equivalences above give the all-open and choice-independent statements. The main theorems are collected in `SheafyEndpoints`, and include the following.

```
theorem finiteJet_isDomain : IsDomain (JetA K) :=
  inferInstance

theorem finiteJet_not_noetherian :
  ¬ IsNoetherianRing (JetA K) :=
  not_isNoetherianRing_JetA K

theorem finiteJet_isUniform_of_dvr
  [IsDiscreteValuationRing  $\mathcal{O}[K]$ ] :
  TopologicalRing.IsUniform (JetA K) :=
  finiteJet_isUniform K (Uniformizer.ofDVR K)

theorem finiteJet_not_stablyUniform_of_dvr
  [IsDiscreteValuationRing  $\mathcal{O}[K]$ ] :
  ¬ TopologicalRing.IsStablyUniform (JetA K) :=
  finiteJet_not_stablyUniform K (Uniformizer.ofDVR K)

theorem finiteJet_isSheafyFor_of_dvr
  [IsDiscreteValuationRing  $\mathcal{O}[K]$ ] :
  IsSheafyFor (JetA K) (finiteJetPlus K) :=
  finiteJet_isSheafyFor K (Uniformizer.ofDVR K)
  (isNoetherianRing_unitBall K)

theorem finiteJet_isSheafyTateRing_of_dvr
  [IsDiscreteValuationRing  $\mathcal{O}[K]$ ] :
  IsSheafyTateRing (JetA K) :=
  finiteJet_isSheafyTateRing K (Uniformizer.ofDVR K)
  (isNoetherianRing_unitBall K)
```

Here  $K$  is any complete nontrivially normed ultrametric field for which  $\mathcal{O}[K] = \{x : \|x\| \leq 1\}$  is a discrete valuation ring. No local compactness, finiteness of the residue field, characteristic, or perfection assumption is used. The theorem `finiteJet_isSheafyFor_of_dvr` proves that  $(A, A^\circ)$  is sheafy, whilst `finiteJet_isSheafyTateRing_of_dvr` removes the choices of completion and ring of integral elements. The corresponding all-open statement is

```
TopCat.Presheaf.IsSheafOfTopologicalRings
  (ValuationSpectrum.structurePresheaf (JetA K)),
```

The corresponding Lean theorem proves this statement directly.

**A.4. Scope of the finite-jet formalisation.** The paper's  $A_0$  is represented by the Gauss-norm unit ball. The theorem `isPowerBounded_JetA_iff` identifies this unit ball with the power-bounded elements, giving  $A^\circ = A_0$ . Completeness and the Tate structure are supplied by the corresponding instances. As in the weighted-parity development, the constant-series map from  $K$  is explicit and isometric but is not installed as an `Algebra K` instance.

For the bad chart, `chartEquiv` is an exported ring equivalence. The two continuity theorems make it a homeomorphism, and the canonical-map theorem identifies  $W$  and  $Q$ . It is not bundled as a `K-AlgEquiv`, which is why proposition 4.1 is stated for complete topological rings.

The conclusions of lemma 5.1 are formalised in the normed setting used for the example:  $E$  is one of  $B, C, D, T_E$  has its Gauss norm,  $T_{E,0}$  is its unit ball, and the finite free terms have their sup norms. Positive-degree exactness is transported from the polynomial sequence by flat base change. Closedness in positive degree comes from exactness and continuity, whilst the degree-zero ideal is closed by the noetherian closed-ideal theorem. The closed-range theorem then gives controlled lifts and strictness; scaling those lifts gives the two displayed denominator inclusions.

The proof of lemma 5.2 in Lean uses these controlled lifts directly. Its localisation proof obtains the topological embedding from a quotient-norm estimate, and proves openness of  $C_\alpha \rightarrow D_\alpha$  from the norm-preserving coefficientwise section. The components of proposition 5.3 are therefore separate declarations. Finally, the development proves the concrete finite-jet transfer `isSheafy_JetA`: it pushes finite rational covers to  $B, C, D$ , glues there, and uses the localised Milnor row and a closed-range theorem.

The Koszul argument, the rational-chart calculation, and the transfer of sheafiness through the concrete Milnor square are contained respectively in the restricted-exactness file and the strictness and closed-image file, `FJP/Over/Chart.lean`, and `FJP/Over/SheafTransfer.lean`.

**A.5. The weighted-parity example.** The second development uses the same restricted-series infrastructure. Its ambient ring is the countable restricted Tate algebra  $K\langle W, U_1, U_2, \dots \rangle$ . The definitions `wpWeight`, `WPMem`, and `wpSupport` are the direct translation of (13)–(17). The relevant part of the definition is as follows.

```
def wpWeight (w : ℕ → ℕ) (t : ℕ →₀ ℕ) : ℕ :=
  ∑ n ∈ t.support,
    if (t n).mod 2 = 1 ∧ n ≠ 0 then w n else 0

def WPMem (w : ℕ → ℕ) (t : ℕ →₀ ℕ) : Prop :=
  wpWeight w t ≤ t 0

abbrev WPA : Type _ := ↑(wpSupport K w)

def idWeight : ℕ → ℕ := fun n => n

abbrev WPAid : Type _ := WPA K idWeight
```

The coordinate  $t(0)$  is the exponent of  $W$ ; the other coordinates are the exponents of the  $U_n$ . Thus `WPMem idWeight t` says exactly that  $a \geq \omega(\nu)$ , and `wpSupport` requires the coefficients outside this monoid to vanish. It follows that `WPAid K` is the ring  $\mathcal{A}$  of (17). The maps `constA` and `norm_constA` give the isometric embedding of  $K$  as the constant series. This map is explicit in the development, although it is not bundled as an `Algebra K` instance.

Suppose that  $K$  is complete and nontrivially normed and that  $\mathcal{O}[K]$  is a DVR. The principal exported statements are:

```

theorem weightedParity_isUniform_of_dvr :
  IsUniform (WPAid K) :=
  weightedParity_isUniform (Uniformizer.ofDVR K)

theorem weightedParity_isDomain :
  IsDomain (WPAid K) :=
  inferInstance

theorem weightedParity_not_noetherian :
  ¬ IsNoetherianRing (WPAid K) :=
  not_isNoetherianRing_WPA K idWeight idWeight_pos

theorem weightedParity_powerBounded_eq_unitBall
  (ϖ : Uniformizer K) :
  powerBoundedSubring (WPAid K) =
  (FiniteJet.unitBall (WPAid K) : Set (WPAid K)) :=
  powerBoundedSubring_eq_unitBall ϖ

theorem weightedParity_isSheafyComplete_of_dvr :
  IsSheafyComplete (WPAid K) :=
  weightedParity_isSheafyComplete (Uniformizer.ofDVR K)
  (FiniteJetOver.isNoetherianRing_unitBall K)

theorem weightedParity_chart_isDomain
  (ϖ : Uniformizer K)
  (hK₀ : IsNoetherianRing (FiniteJet.unitBall K)) :
  IsDomain (presheafValue
    (chartDatum (w := idWeight) ϖ)) :=
  isDomain_chart ϖ hK₀

theorem weightedParity_chart_not_isUniform
  (ϖ : Uniformizer K)
  (hK₀ : IsNoetherianRing (FiniteJet.unitBall K)) :
  ¬ IsUniform (presheafValue
    (chartDatum (w := idWeight) ϖ)) :=
  not_isUniform_chart idWeight_unbounded ϖ hK₀

theorem weightedParity_not_stablyUniform_of_dvr :
  ¬ IsStablyUniform (WPAid K) :=
  weightedParity_not_stablyUniform (Uniformizer.ofDVR K)
  (FiniteJetOver.isNoetherianRing_unitBall K)

```

Here `IsSheafyComplete` is the predicate defined above: it quantifies over every ring of integral elements. In particular it proves the sheaf condition for  $(\mathcal{A}, \mathcal{A}^\circ)$ . The all-open theorem also records the corresponding statement directly for the structure presheaf. The datum

`chartDatum` is  $(W; \varpi)$ , and `chartDatum_isRational` checks that it defines the chart  $|W| \leq |\varpi|$ . The two chart theorems then say precisely that its section ring is a domain but is not uniform.

The proof routes are the ones used in Section 7. Multiplicativity of the countable Gauss norm gives the domain and unit-ball statements. Finite-head localisation is proved by taking the completed head quotient, applying its evaluation map coefficientwise to the tail, and constructing the inverse by summing a  $c_0$ -family; it does not invoke the Koszul lemma. For sheafiness, one common head is chosen for the base datum and all members of a finite cover. Head sheafiness glues each coefficient, and the head restriction embedding shows that the resulting coefficients tend to zero. The final topological embedding is obtained from the closed matching equaliser by the Tate closed-range theorem. For the bad chart, the head quotient is  $K\langle X \rangle$  and the tail algebra embeds in formal multivariable power series over this domain. The small-perturbation lemma of Section 7 is the unit-ball, norm-scaling theorem formalised by `WeightedParity.PerturbSetup`.

These declarations, including the final failure of stable uniformity, have been checked to use only the usual logical axioms `propext`, `Classical.choice`, and `Quot.sound`.

**A.6. Mathematical and code provenance.** The formal structure presheaf, rational-cover criterion, and the strongly-noetherian sheaf theorem follow Huber’s construction [7, Theorem 2.2 and Lemmas 2.3–2.6] and Wedhorn’s systematic formulation [16, Proposition and Definition 6.36, Remark 8.20, Definition 8.26, Theorem 8.28(b), Corollary 8.32, and Lemmas 8.33–8.34]. The passage between a fixed plus ring and all plus rings follows the standard rational-refinement argument recorded by Kedlaya [8, Lemma 1.6.8 and Remark 1.6.9]. Ben-Bassat–Kremnizer prove the strict one-variable resolutions used in their treatment of rational localisations [2, Lemmas 5.13–5.14]; Bambozzi–Kremnizer formulate the general construction in terms of Koszul complexes and prove strict Koszul regularity [3, Notation 4.3, Definitions 4.5–4.6, and Corollary 4.13]. The formalisation does not invoke either result. Positive-degree exactness is instead proved by flat base change from the polynomial sequence; closedness and the nonarchimedean open mapping theorem [9, Lemma 3.1] give controlled lifts. Scaling those lifts yields the bounded-denominator criterion used by Buzzard–Verberkmoes [4, Lemma 2].

The development is written in Lean 4 [10] on top of `mathlib` [11]. The formalisation-specific files live in `AINTLIB` [1]. Its restricted power-series and restricted Laurent-series infrastructure includes Apache-2.0-licensed code adapted from William Coram’s PhD repository, with equivalence code attributed there to Bingyu Xia [5]. The vendored source headers and commit history retain those attributions. The finite-jet and weighted-parity definitions, localisation proofs, and sheaf endpoints discussed in this appendix are the `AINTLIB`-specific part of the development.

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